# ETALENESS OF A BASE-CHANGED ALGEBRA MAP IS REALISED AT A FINITE STAGE OF THE COLIMIT

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*Allowed standard prerequisites (editorial).*

- Etaleness tested by the differential of a presentation. Used at proof lines 47864-47871. Turns the hypothesis into the statement that the map from the relation module modulo squares to the free module of differentials is an isomorphism.
- Commutation of tensor products with directed colimits. Used when descending the finitely many elements to a stage (lines 47871-47878). Moves the chosen preimages of the differentials and their relations down to a single index.

**Lemma 0.1.** *Let $A = \operatorname{colim}_{i \in I} A_i$ be a directed colimit of rings. Let $0 \in I$ and $\varphi_0 : B_0 \to C_0$ a map of $A_0$-algebras. Assume*

(1) *$A \otimes_{A_0} B_0 \to A \otimes_{A_0} C_0$ is étale,*
(2) *$B_0 \to C_0$ is of finite presentation.*

*Then for some $i \geq 0$ the map $A_i \otimes_{A_0} B_0 \to A_i \otimes_{A_0} C_0$ is étale.*

*Proof.* Write $C_0 = B_0[x_1, \ldots, x_n]/(f_{1,0}, \ldots, f_{m,0})$. Write $B_i = A_i \otimes_{A_0} B_0$ and $C_i = A_i \otimes_{A_0} C_0$. Note that $C_i = B_i[x_1, \ldots, x_n]/(f_{1,i}, \ldots, f_{m,i})$ where $f_{j,i}$ is the image of $f_{j,0}$ in the polynomial ring over $B_i$. Write $B = A \otimes_{A_0} B_0$ and $C = A \otimes_{A_0} C_0$.

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Note that $C = B[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$ where $f_j$ is the image of $f_{j,0}$ in the polynomial ring over $B$. The assumption is that the map

$$\mathrm{d} : (f_1, \ldots, f_m)/(f_1, \ldots, f_m)^2 \longrightarrow \bigoplus C\mathrm{d}x_k$$

is an isomorphism. Thus for sufficiently large $i$ we can find elements

$$\xi_{k,i} \in (f_{1,i}, \ldots, f_{m,i})/(f_{1,i}, \ldots, f_{m,i})^2$$

with $\mathrm{d}\xi_{k,i} = \mathrm{d}x_k$ in $\bigoplus C_i\mathrm{d}x_k$. Moreover, on increasing $i$ if necessary, we see that $\sum(\partial f_{j,i}/\partial x_k)\xi_{k,i} = f_{j,i} \bmod (f_{1,i}, \ldots, f_{m,i})^2$ since this is true in the limit. Then this $i$ works. $\square$